

EXP NO:14

Determination and plotting of Polarization in different Dielectric Medium with their relative permittivity and to find electric susceptibility.

Enter the ϵ_r value: 3

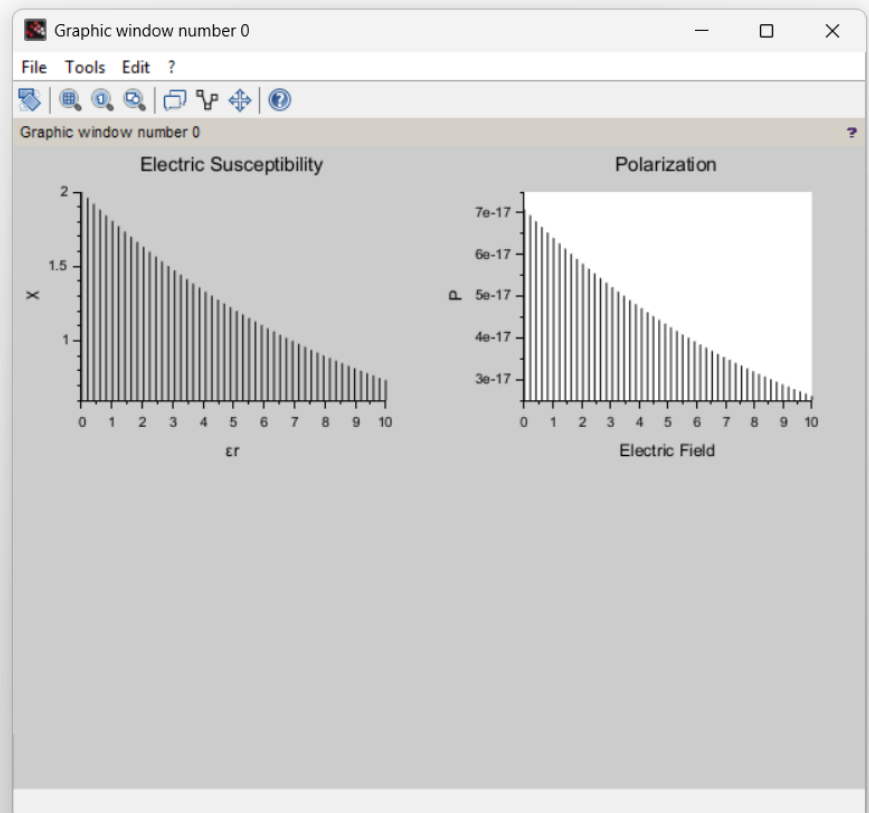
Enter the Electric Field: 4×10^{-6}

"Electric susceptibility $\chi = 2$ "

"Polarization $P = 7.083D-17$ "

```
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci)...
File Edit Format Options Window Execute ?
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
*exp333.sci
1 clc;
2 clear;
3 epsilon = -8.854e-12;
4 er = input("Enter the er value: ");
5 field = input("Enter the Electric Field: ");
6 chi = er - 1;
7 polarization = chi * epsilon * field;
8 disp("Electric susceptibility  $\chi = "$  + string(chi));
9 disp("Polarization P = " + string(polarization));
10 x = linspace(0,10,50);
11 y = chi * exp(-0.1*x);
12 z = polarization * exp(-0.1*x);
13 figure();
14 subplot(2,2,1);
15 plot2d3(x,y);
16 title("Electric-Susceptibility");
17 xlabel("er");
18 ylabel("χ");
19 subplot(2,2,2);
20 plot2d3(x,z);
21 title("Polarization");
22 xlabel("Electric-Field");
23 ylabel("P");
24
```

Line 18, Column 12.



Calculation:

$$\epsilon_r = 14$$

Enter the $\epsilon_r = 3$

Enter Electric Field = 4×10^{-6}

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

i) Electric susceptibility:

$$\chi = \epsilon_r - 1$$

$$\chi = 3 - 1$$

$$\chi = 2 //$$

(ii) Polarization

$$\vec{P} = \chi \epsilon_0 \vec{E}$$

$$P = 2 (8.854 \times 10^{-12}) \times (4 \times 10^{-6})$$

$$2 \times 8.854 \times 10^{-12} = 17.708 \times 10^{-12}$$

$$P = (17.708 \times 10^{-12}) (4 \times 10^{-6})$$

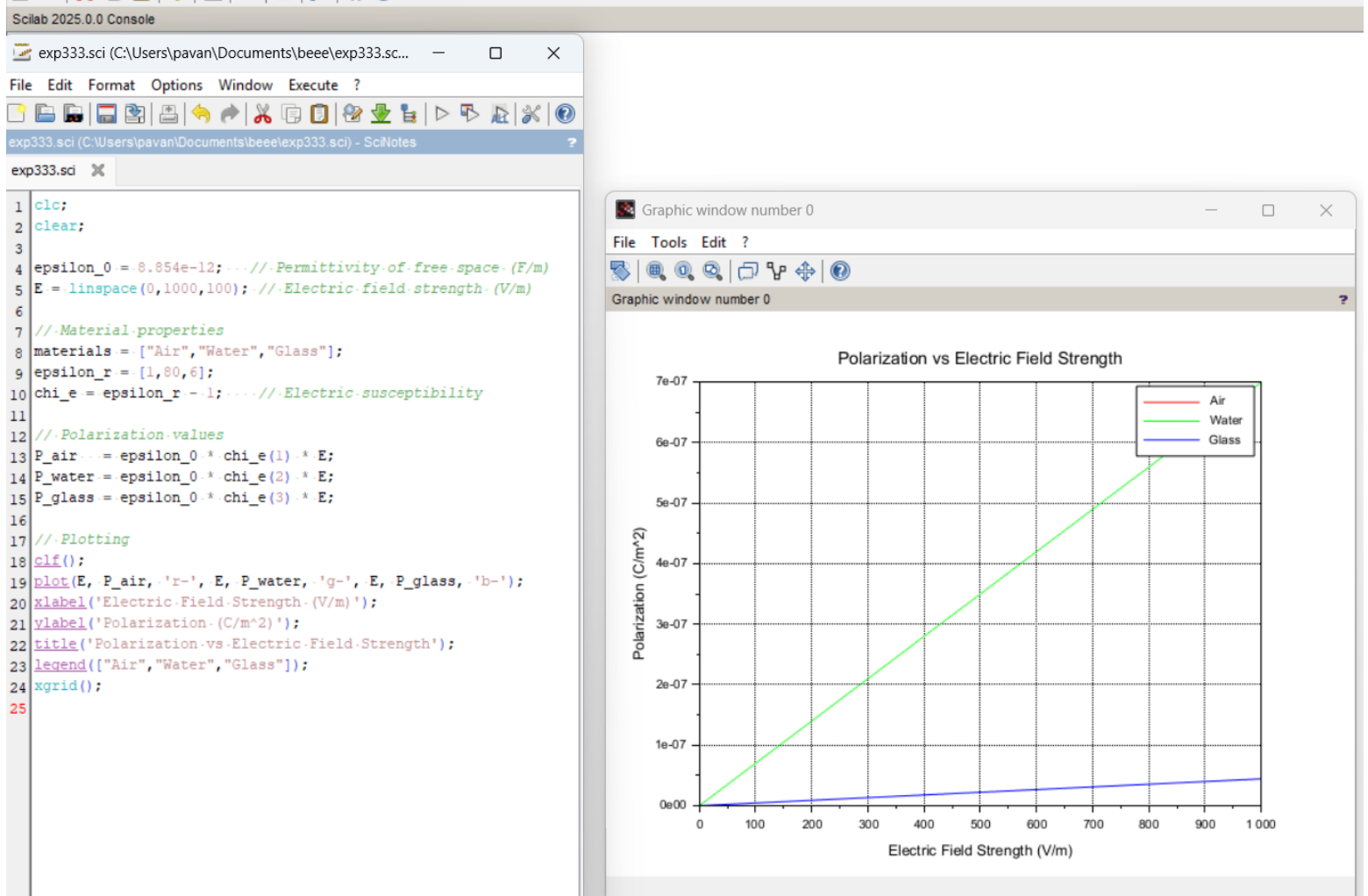
$$P = 70.832 \times 10^{-18}$$

$$P = 7.083 \times 10^{-17} \text{ C/m}^2$$

$$P = 7.083 \text{ D-17} //$$

CASE:1

To determine and plot the polarization and electric susceptibility of electromagnetic waves in three dielectric media: Air, Water, and Glass, using their relative permittivity.

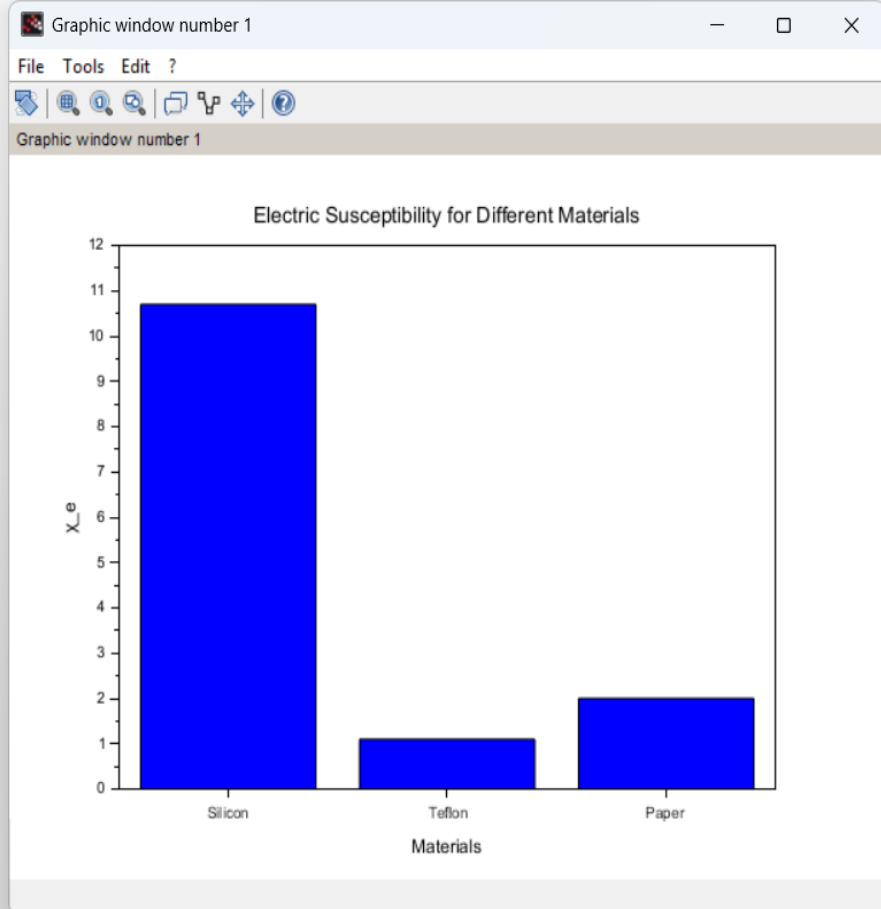


CASE:2 To calculate and plot the polarization and electric susceptibility in Silicon, Teflon, and Paper, and understand how their relative permittivity affects the polarization.

"Material | Relative Permittivity | Electric Susceptibility (χ_e) | Polarization (P in C/m²)"
at line 25 of executed file C:\Users\pavan\Documents\beee\exp333.sci

!': Wrong type for argument 2: Real scalar expected.

```
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sc...
File Edit Format Options Window Execute ?
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
*exp333.sci
1 clc;
2 clear;
3 epsilon0 = 8.854e-12; //Permittivity of free space (F/m)
4 materials = ["Silicon", "Teflon", "Paper"];
5 epsilon_r = [11.7, 2.1, 3.0]; //Relative permittivity
6 chi_e = epsilon_r - 1; //Electric susceptibility
7 E = 1e6; //Applied Electric Field in V/m
8 polarization = epsilon0 * chi_e * E; //Polarization (C/m^2)
9 scf(1);
10 clf(1);
11 bar(chi_e);
12 xtitle("Electric Susceptibility for Different Materials", "Mat
   erials", "χe");
13 gca().x_ticks = tlist(["ticks", "locations", "labels"], [1, 2,
   3], materials);
14 disp("Material | Relative Permittivity | Electric Susceptibili
   ty (χe) | Polarization (P in C/m^2)");
15 for i = 1:length(materials)
16     disp(materials(i) + " | " + string(epsilon_r(i)) + " | " +
   string(chi_e(i)) + " | " + string(polarization(i)));
17 end
18 scf(2);
19 clf(2);
20 bar(polarization);
21 xtitle("Polarization for Different Materials", "Materials", "P
   olarization (P) in C/m^2");
22 gca().x_ticks = tlist(["ticks", "locations", "labels"], [1, 2,
   3], materials);
23
```



case 3: To explore how different dielectric materials (Rubber, Oil, and Plastic) affect the polarization and electric susceptibility, and to compare their relative permittivity's.

"Material | Relative Permittivity | Electric Susceptibility (χ_e) | Polarization (P in C/m²)"
at line 25 of executed file C:\Users\pavan\Documents\beee\exp333.sci

':': Wrong type for argument 2: Real scalar expected.

```
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
File Edit Format Options Window Execute ?
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
*exp333.sci X
1 clc;
2 clear;
3 epsilon0 = -8.854e-12; // Permittivity of free space (F/m)
4 materials = ["Rubber", "Oil", "Plastic"];
5 epsilon_r = [3.5, 2.2, 2.3]; // Relative permittivity
6 chi_e = epsilon_r - 1; // Electric susceptibility
7 E = 1e6; // Applied Electric Field in V/m (example value)
8 polarization = epsilon0 * chi_e * E; // Polarization (C/m^2)
9 scf(1);
10 clf(1);
11 bar(chi_e); // Plot electric susceptibility as a bar chart
12 xtitle("Electric Susceptibility for Different Materials", "Materials", "chi_e");
13 gca().x_ticks = tlist(["ticks", "locations", "labels"], [1, 2, 3], materials);
14 disp("Material | Relative Permittivity | Electric Susceptibility (chi_e) | Polarization (P in C/m^2)");
15 for i = 1:length(materials)
16     disp(materials(i) + " | " + string(epsilon_r(i)) + " | " + string(chi_e(i)) + " | " + string(polarization(i)));
17 end
18 scf(2);
19 clf(2);
20 bar(polarization);
21 xtitle("Polarization for Different Materials", "Materials", "Polarization (P) in C/m^2");
22 gca().x_ticks = tlist(["ticks", "locations", "labels"], [1, 2, 3], materials);
23
```

